

Team Details

(Team : ***Solo_Levlling***)



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Problem Statement :

The Triage Bottleneck

- ▶ **Clinical Reality** -- When emergency departments are overwhelmed, patient sequencing relies heavily on rapid human judgment under extreme pressure. A single triage decision can determine whether a deteriorating patient is seen immediately or left waiting, while simultaneously determining how scarce clinical resources are utilized.
- ▶ **Main Focus** -- The triage nurse must answer two fundamentally different questions at once: “**Who needs human attention first?**” and “**What can safely happen right now?**”. Treating the ED as a single priority queue creates unnecessary waiting when diagnostic or preparatory actions could happen in parallel.
- ▶ **Worst Case** -- The system should be optimized for **safe under-triage avoidance**, not average-case accuracy. When early data is ambiguous, subtle indicators of deterioration trigger escalation rather than silent de-prioritisation. Every recommendation must remain bounded by clinical oversight.

Solution Design :

The Intelligent ED Co-Pilot

PatientTriage.ai is an intelligent co-pilot that separates Clinical Urgency from Operational Flow. Rather than relying on a black-box machine learning model that introduces hallucination risks, the solution utilizes a Hybrid-AI architecture:

- ▶ - **Fuzzy NLP Parsing** : An LLM normalizes raw, typo-ridden triage notes and shorthand into standardized clinical tokens.
- ▶ - **Safety Agent (Clinical Urgency)** : A deterministic, mathematically explainable expert system processes the first 3-5 minutes of fragmented data (vitals, demographics, medical history) to assess patient risk. It uses age-adjusted thresholds and is explicitly optimized for safe under-triage avoidance—biasing toward escalation when early data is ambiguous.
- ▶ - **Flow Agent (Operational Flow)** : Simultaneously predicts and schedules what actions can safely occur in parallel, recommending the patient's likely next resources (e.g., ECG, X-ray, blood tests).

Target Users

The platform is designed for three distinct tiers of ED stakeholders:

- **Primary Users (Triage Nurses)** : Benefit from reduced cognitive burden. The system acts as a safety net, automatically flagging deteriorating patients and allowing nurses to confidently manage surge queues.
- **Secondary Users (ED Charge Nurses & Attending Physicians)** : Benefit from the Flow Agent's parallel scheduling, allowing clinical teams to prep beds and order labs before the physician even sees the patient, accelerating time-to-treatment.
- **Tertiary Users (Hospital Administrators & Compliance Officers)** : Benefit from the strict audit trails. Every clinical override is permanently logged with mandatory justifications, ensuring total HIPAA compliance and legal accountability.



Business Case and Impact

The financial and operational impacts of PatientTriage.ai are significant.



Increased ED Throughput: By moving away from a single priority queue and parallelizing diagnostic workflows, hospitals can decrease the average Length of Stay (LOS) in the ED, allowing them to treat a higher daily volume of patients.



Risk Mitigation & Liability Reduction: Missing a critical case carries massive asymmetry in healthcare costs (malpractice suits, intensive care escalation). By mathematically preventing silent de-prioritization and enforcing an immutable audit trail for manual overrides, the hospital significantly reduces its liability footprint.



Resource Optimization: Efficiently routing patients prevents high-acuity beds from being occupied by low-acuity patients awaiting basic lab work.

Phased Roadmap

To scale from a hackathon prototype to a fully integrated enterprise health platform, I propose the following phased rollout:



Phase 1: Retrospective Validation (Months 1-2):

Run anonymized historical ED patient data through the Safety Agent. Compare the deterministic AI's triage scoring against the historical human decisions to validate safety and under-triage avoidance.



Phase 2: Shadow Deployment & Pilot (Months 3-5):

Deploy the dashboard in a high-volume, localized setting (such as piloting at the IIT Delhi campus hospital or a partner NCR medical facility). The system runs in "shadow mode" alongside triage nurses to gather real-world usability feedback without driving final clinical decisions.



Phase 3: EHR Integration (Months 6-8): Build secure, bi-directional API pipelines with major Electronic Health Record (EHR) providers (Epic, Cerner) to automate the MRN lookup process and pull historical medical records instantly.



Phase 4: Live Rollout & Multi-Hospital Scaling (Months 9+):

Full deployment with active Flow Agent resource scheduling. The deterministic rules-engine allows the system to scale easily across hospitals of varying sizes and specialty mixes without requiring expensive localized ML retraining.

Key Risks and Mitigations

Identified Risk	Impact	Mitigation Strategy
"Black-Box" AI Hallucinations	High (Patient Safety)	Hybrid Architecture: The LLM is restricted exclusively to text normalization. All clinical risk stratification is strictly handled by a deterministic, transparent rules engine bounded by clinical oversight.
Clinician Resistance & Mistrust	High (Adoption)	Human-in-the-Loop Design: The tool is positioned as a "co-pilot," not a replacement. The UI includes a prominent "Override" mechanism, empowering fatigued nurses to exercise and enforce their clinical judgment at any time.
Data Privacy & Compliance Breaches	Severe (Legal/Financial)	Immutable Audit Trails: Built on a robust PostgreSQL/Prisma relational architecture. The system utilizes atomic transactions to ensure no AI score can be manually altered without securely logging the clinician ID, role, and text justification.